

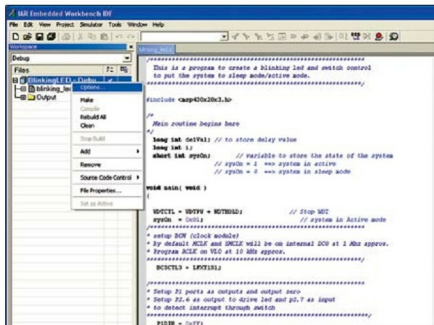


Fig. 12: Screenshot of the BlinkingLED project

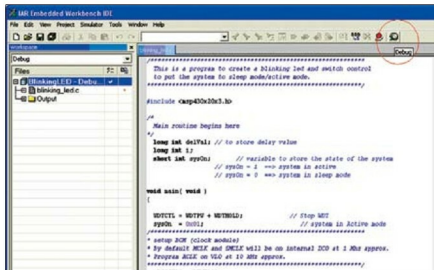


Fig. 13: Screenshot of the debug icon used for debugging and downloading the code into target board

board should work off the external 3V battery.

After successfully testing the LED blinking application program, you can proceed to the digital thermometer application. Repeat the above steps for digital thermometer program.

## Designing a low-cost precision thermometer

Here we describe a simple add-on circuit for a digital multimeter that enables direct read-out of temperature

with fairly high degree of accuracy. Designing with MSP430 microcontroller results in low cost, low power consumption and high precision of the application.

The circuit is designed for use with a commonly available 82-kilo-ohm thermistor bead. A digital voltmeter is connected to the board to read out the temperature directly. For example, a value of 10°C is displayed as 100 mV, 25.6°C is displayed as 256 mV and so on.

The CPU itself works off an internal oscillator, so there is no need to connect an external crystal. The high-resistance (82-kilo-ohm) thermistor keeps the system power consumption low and also makes the error quite negligible. It is connected to 1.2V (Vref) internal voltage reference pin 5 (p1.3) through resistor R27 (680-kilo-ohm). We recommend use of a metal-film resistor for linearity and precise output. Metal-film resistor has better temperature characteristics and current-noise suppression level. It is good to measure the value of this resistor and note down before soldering it, as this value can be used in the program to improve accuracy.

Capacitor C7 (470nF) is used to filter noise on reference voltage pin 5. The voltage across the thermistor is measured using the differential voltage on pins A0+ and A0- of ADC channel 0. This method of measurement also helps in rejecting any common-mode noise that the thermistor leads might pick up.

The temperature output is available at pin 4 of the MSP430 MCU. It is fed to the digital multimeter through an R-C filter formed by resistor R28 and capacitor C6.

The JTAG pin connections are brought out to a 14-pin FRC connector which is used for flash programming and debugging. Push-to-on switch S1 is used to generate a trigger to the system to toggle the system state between active and power-down modes.

Thermistors are highly non-linear components whose resistance decreases with increase in temperature. It has a negative temperature coefficient. Thermistors are good in that these are very sensitive to temperature changes and quite inexpensive. However, their resistance vs temperature characteristic is highly non-linear. Hence normally it requires complex signal conditioning to measure the temperature accurately. It is exactly in this respect that the circuit becomes versatile as the non-linearity of the thermistor behaviour can be easily compensated